

9/4

The 9/4 Matrix Breakthrough

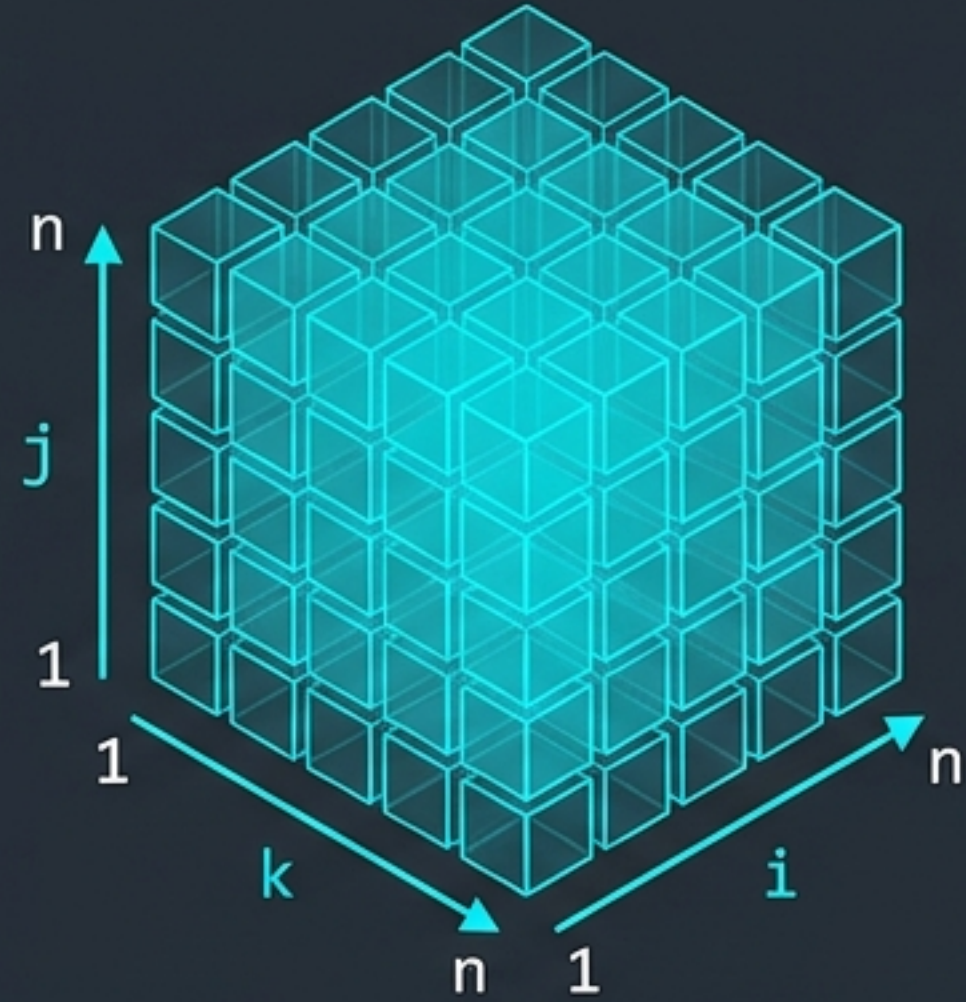
An architectural breakdown of OpenAI's October 2026
proof that $\omega \leq 2.25$ over the complex numbers.

The schoolbook baseline is inherently cubic

THE CODE

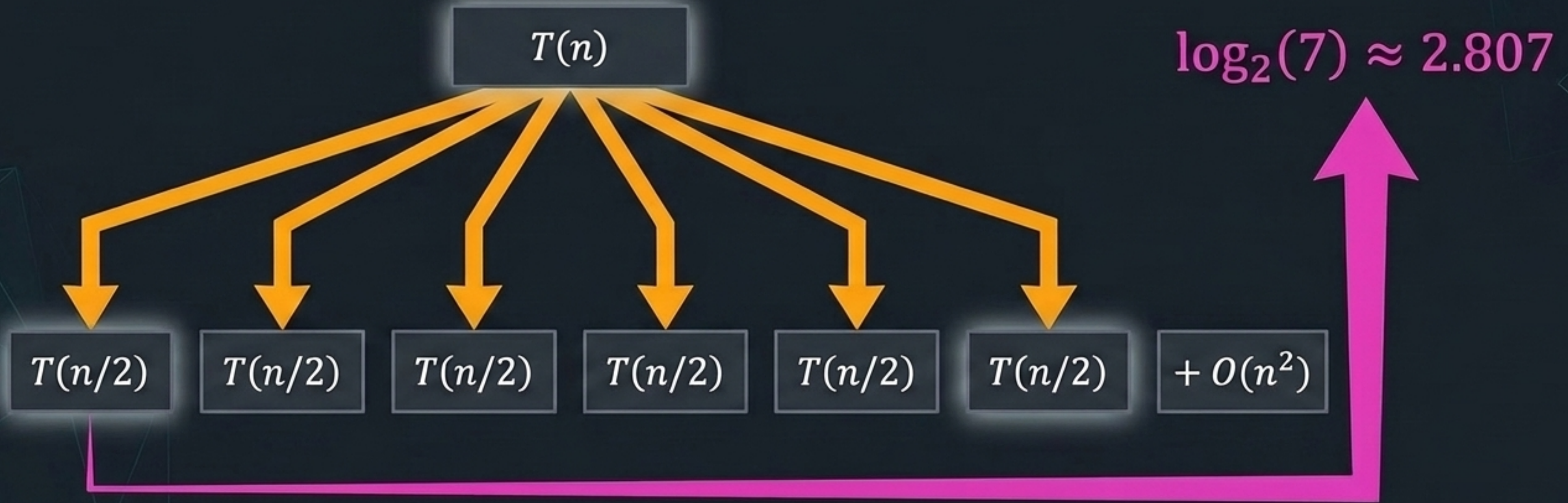
```
for i from 1 to n:  
  for j from 1 to n:  
    for k from 1 to n:  
      C[i][j] = C[i][j] + A[i][k] * B[k][j]
```

THE GEOMETRY



- Every algorithm needs at least $O(n^2)$ steps just to read the two input matrices.
- The standard mathematical definition computes the dot product of rows and columns, forcing an $O(n^3)$ operation.

Strassen's 1969 trick broke the n^3 barrier



- Volker Strassen multiplied 2x2 block matrices using 7 block multiplications instead of 8.

- Applying this recursively yields the equation $T(n) = 7T(n/2) + O(n^2)$.

- This mathematical trick formally drops the runtime to $O(n^{2.807})$.

The ultimate hunt for the exponent ω



The Definition

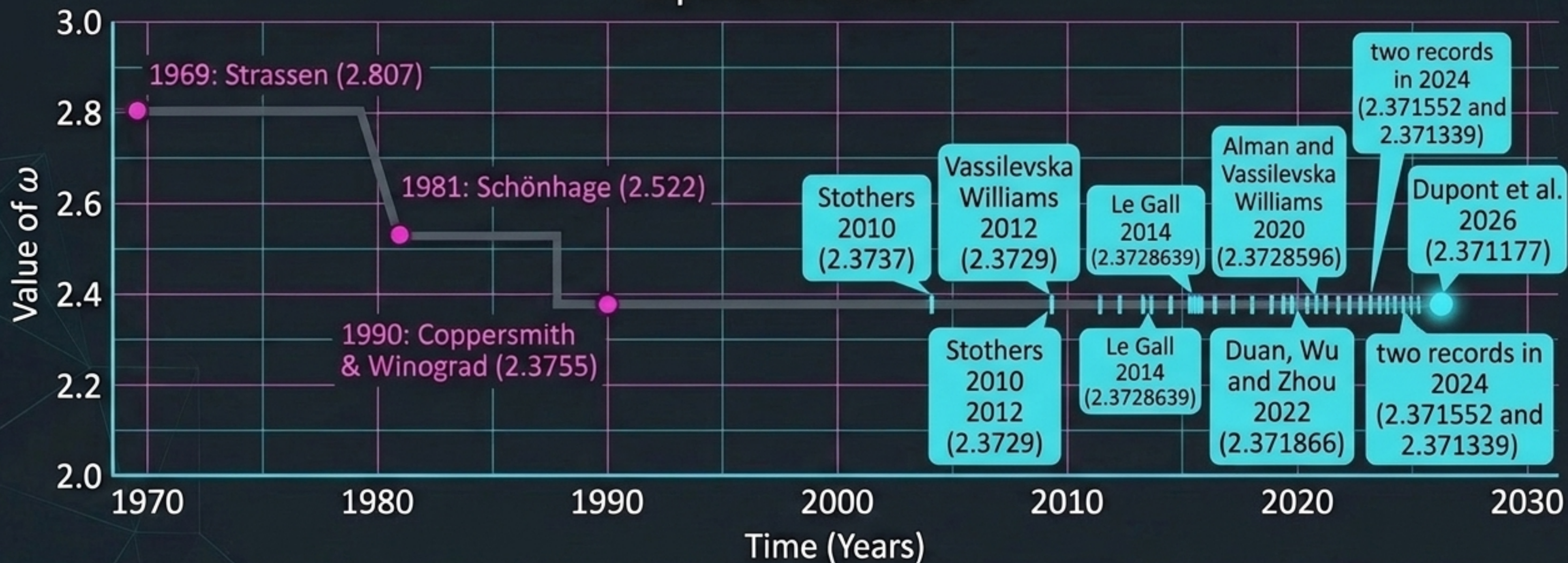
- The exponent ω is the infimum of exponents τ such that $n \times n$ matrices can be multiplied in $O(n^{\tau+\epsilon})$ arithmetic operations.

The Boundaries

- ω must lie somewhere between 2 (the absolute minimum time required to read the input) and 3 (the naive calculation).

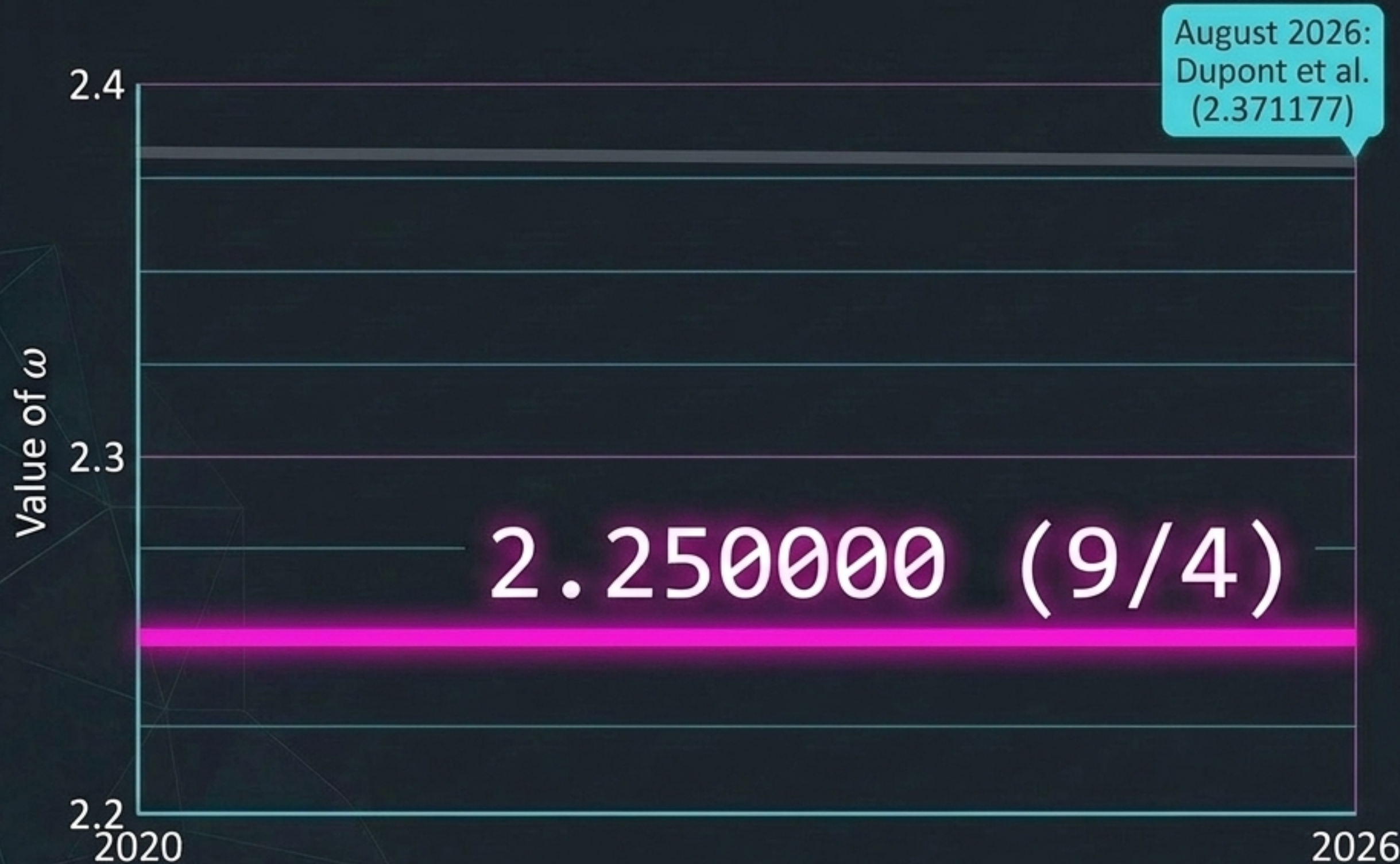
Three decades stuck on the 2.37 plateau

Exponent Staircase



- Strassen introduced the laser method in 1986; Coppersmith and Winograd used it with progression-free sets in 1990 to reach 2.3755, and later records changed only the third decimal place.

OpenAI shatters the plateau with a bound of 2.25



Date:
October 2, 2026

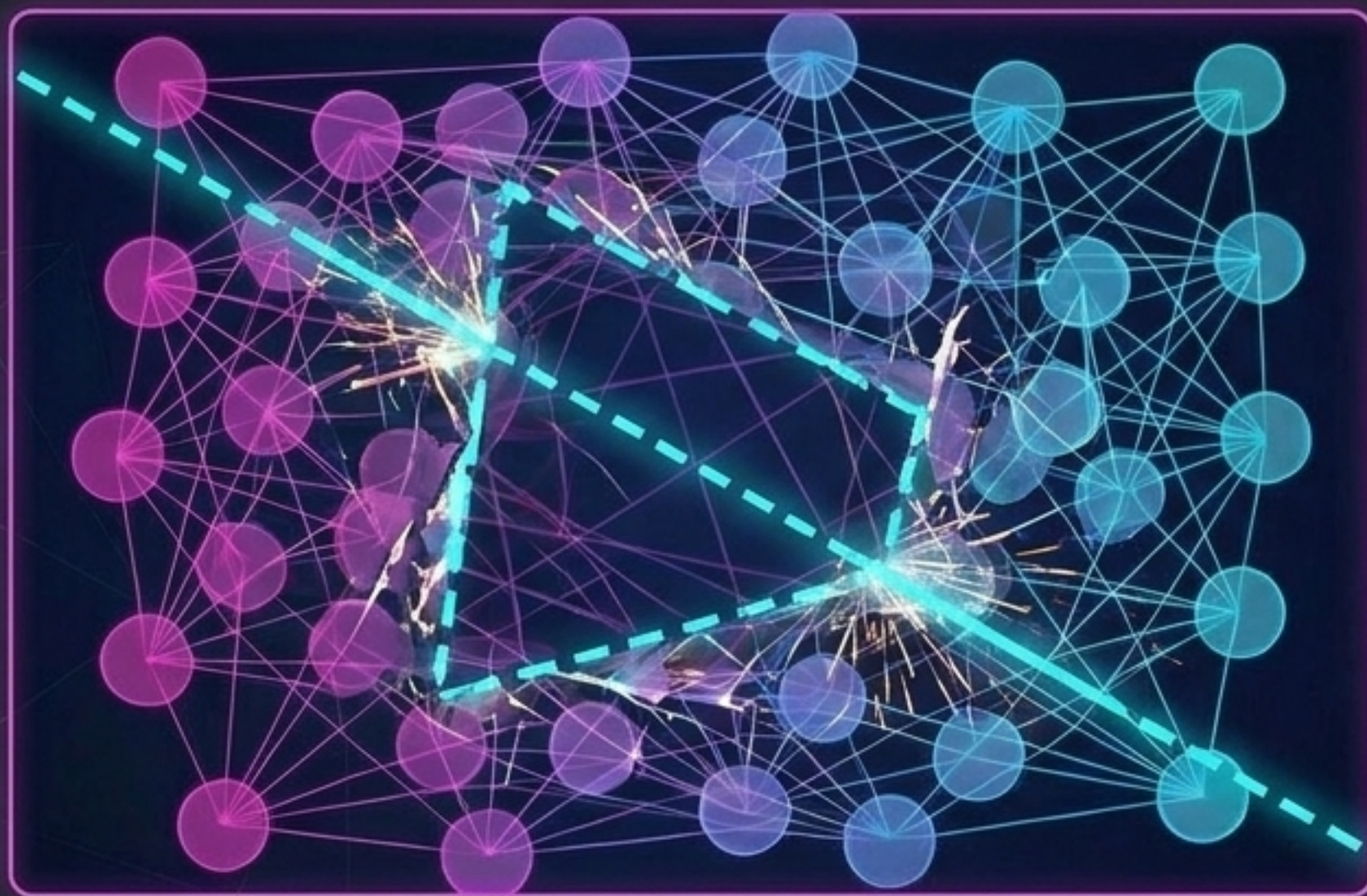
Theorem:
Two $n \times n$ complex
matrices can be
multiplied using
 $O_\varepsilon(n^{9/4+\varepsilon})$ arithmetic
operations.

Result:

$$\omega \leq \frac{9}{4}$$

Abandoning the Laser Method for Tensor Characters

The Old Paradigm (Laser Method)



The New Paradigm (Spectral Viewpoint)



- Previous records relied on Strassen's laser method to forcefully extract independent products from entangled tensor powers. The $9/4$ proof introduces a new framework relying on the theory of **tensor characters**: numerical invariants that mathematically rate tensors without entanglement.

Math Engine Part 1: Rating the Matrices

Function	Add on direct sums, multiply on tensor products.
Dot Product Rating	A length- m dot product is rated precisely as m^p .
Matrix Rating	An $n \times n$ matrix multiplication is rated as n^{3t} .

Tensor characters act as an inescapable rating system. To bound ω , the proof only needs to find the maximum value of the parameter t across all possible characters. If $t \leq 3/4$, then $\omega \leq 9/4$.

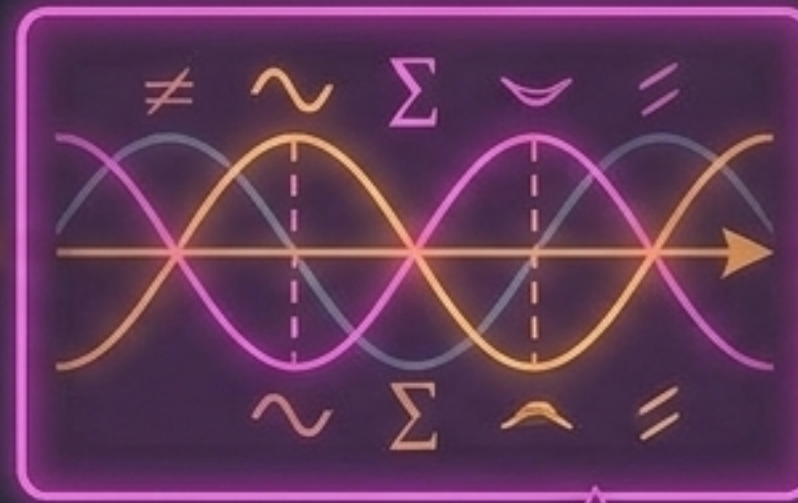
Math Engine Part 2: The Separation Construction

Step 1 (Input)



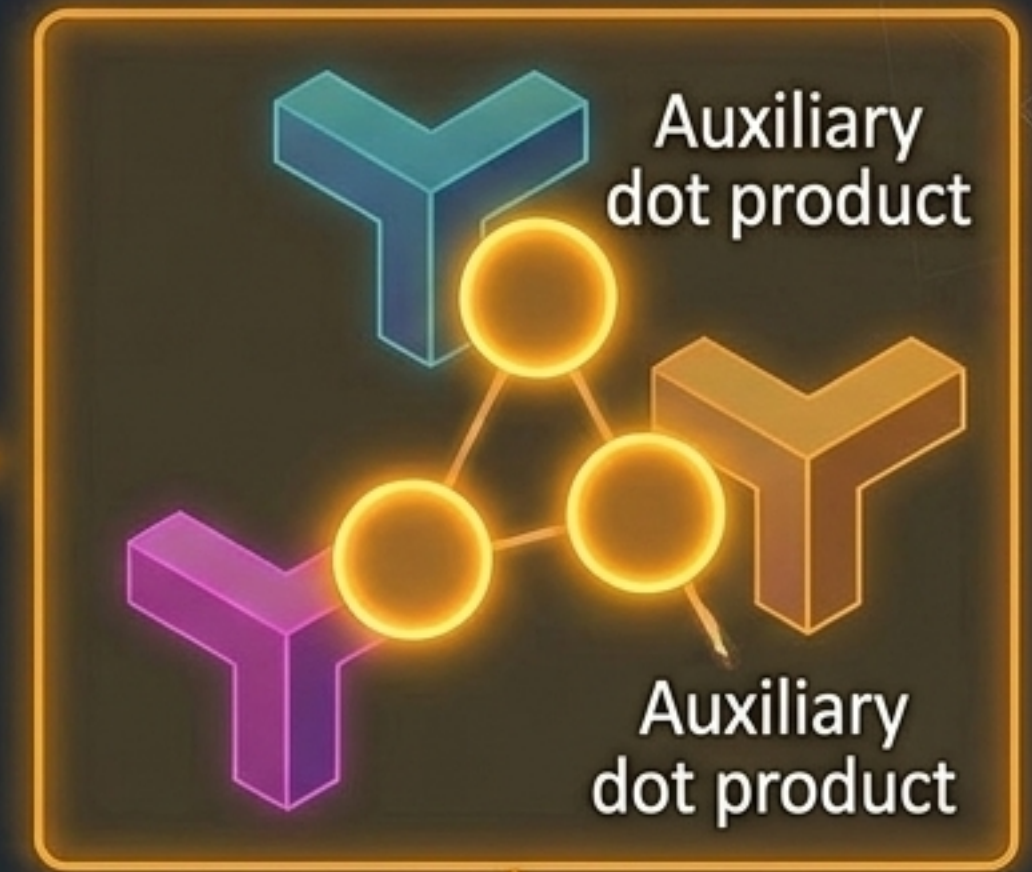
Entangled Blocks
(Shared Leg)

Roots-of-Unity Filter



Applies weights
totaling $(g-h)^2$

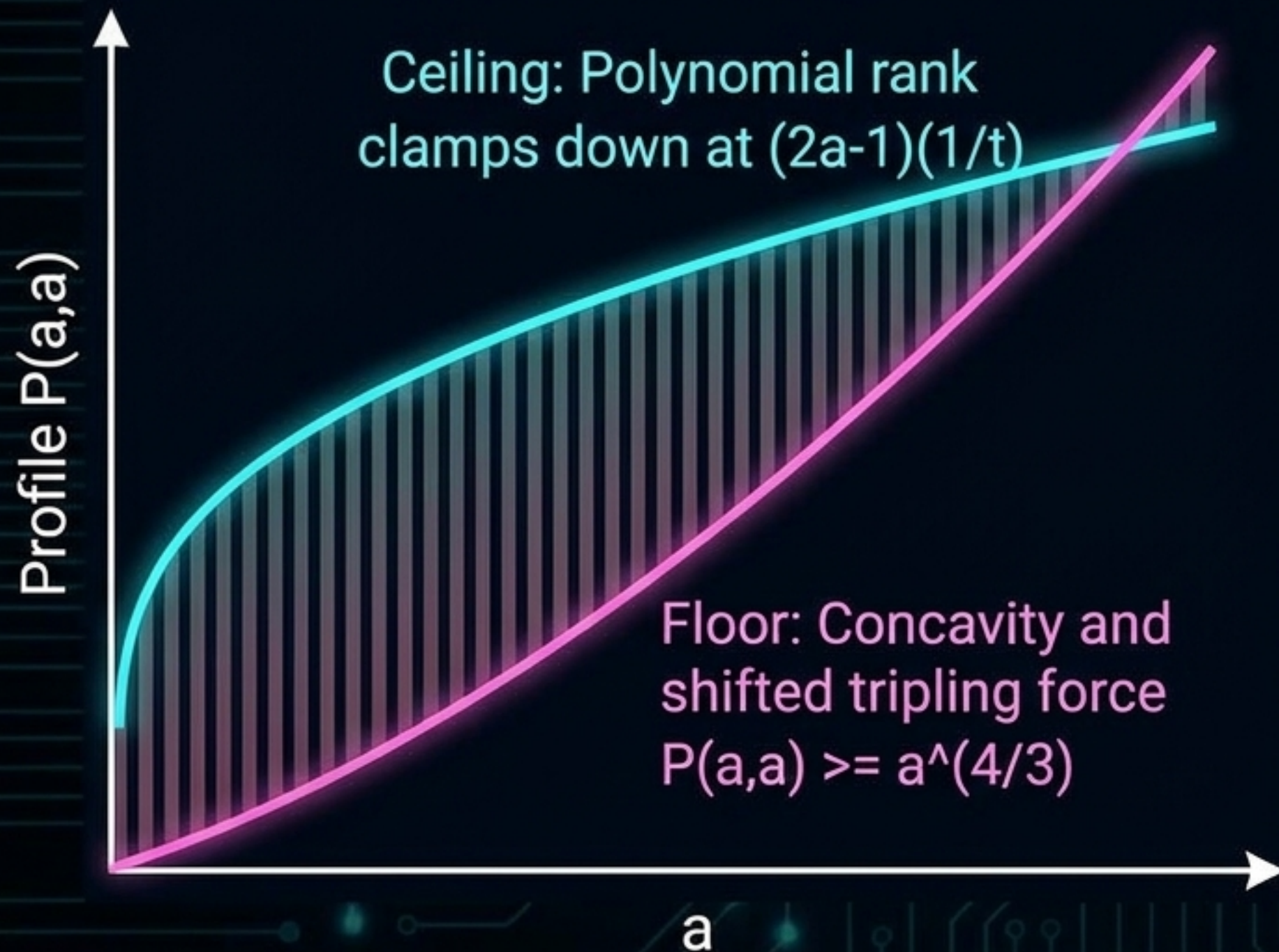
Step 3 (Output)



Independent Blocks
(Untangled)

The core finite construction of the paper. It turns blocks that share a single leg into fully independent blocks, preventing variables from tangling at scale.

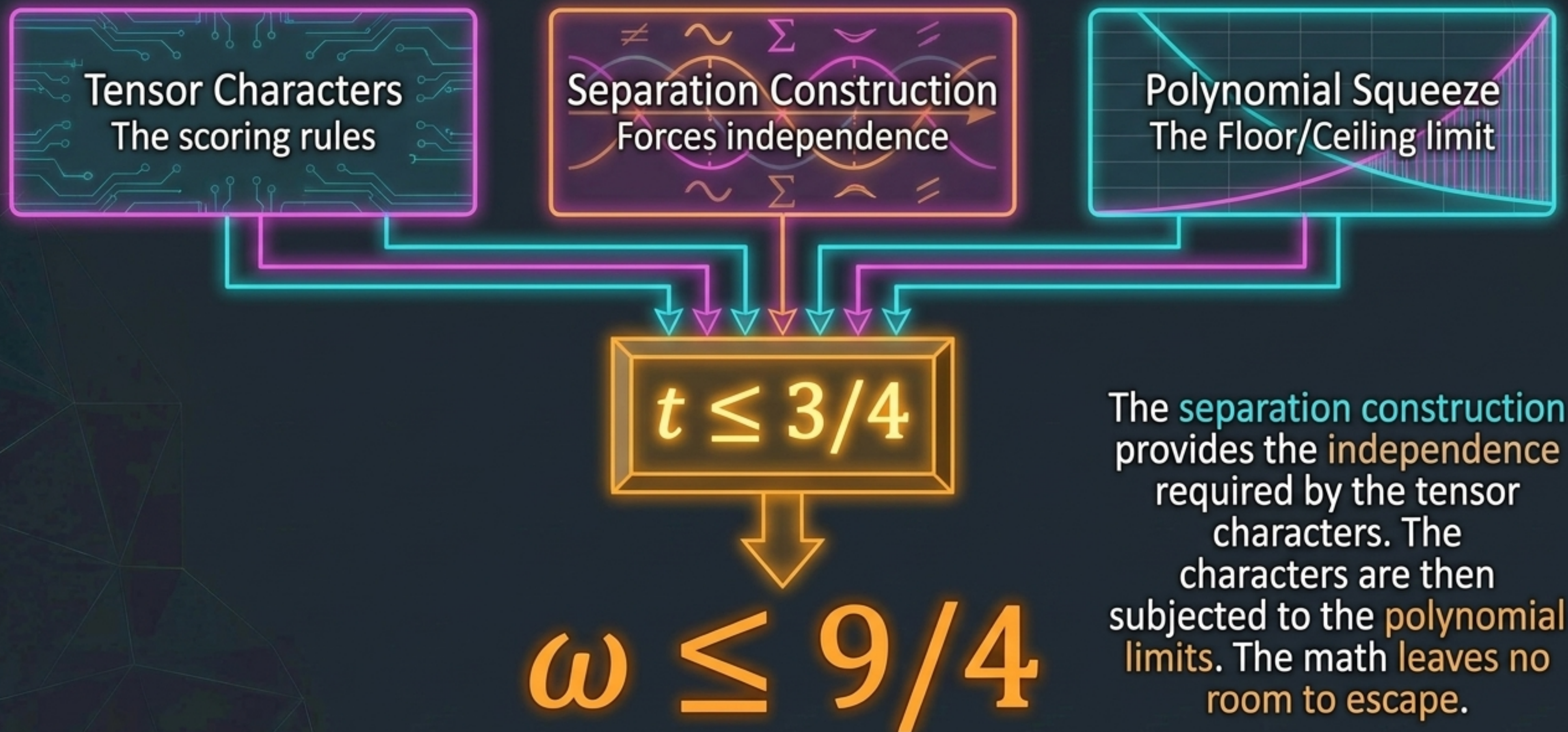
Math Engine Part 3: The Polynomial Squeeze



Polynomial multiplication
creates an unavoidable trap.

Because the $a^{4/3}$ floor
must physically exist below
the ceiling for every value of
 a , the exponent parameter t
is relentlessly squeezed.

The Anatomy of a Bound: How it all converges



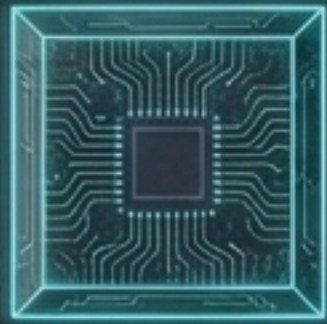
Verification status

```
1 | theorem complex_omega_le_nine_quarters :  
   Arithmetic.omega  $\mathbb{C} \leq (9 : \mathbb{R}) / 4$ 
```

1. The paper is a **preprint** produced by an internal **OpenAI model** and has not yet been **peer reviewed**.
2. The openai/math Lean scope document lists $\omega(\mathbb{C}) \leq 9/4$ as **formalized** in Lean 4, in a model of **finite division-free** arithmetic programs.
3. The openai/math formalization catalogue records its review status as **unchecked**.

The Programmer's Reality Check: Galactic Algorithms

$$O(n^{2.25})$$



$$C \cdot n^{2.25}$$

- The 9/4 result is an **existence proof** about the **asymptotic exponent**, not a ready-to-run algorithm.
- Big-O notation hides a **constant factor**; the paper proves that a **suitable block size exists without naming it**, so the constant and the **crossover size** are unknown.
- The bound is $O(n^{9/4+\epsilon})$ for every $\epsilon > 0$.

Evaluating the landscape of matrix multiplication

Algorithm	Asymptotic Bound (O)	Approach	Practical for Code?
Schoolbook	$O(n^3)$	Nested Loops	Yes (Standard)
Strassen	$O(n^{2.807})$	Recursive 2x2 Blocks	Yes (For large n)
Coppersmith-Winograd	$O(n^{2.375})$	Laser Method	No (Galactic)
OpenAI '26	$O(n^{2.25})$	Tensor Characters	No (Galactic)

The final frontier remains open



- We have shattered the 2.37 plateau and proven $\omega \leq 2.25$.
- However, the ultimate question in theoretical computer science remains unsolved: Can matrices be multiplied in strictly quadratic time?
- $\omega = 2$ remains open.